

Vivek Vikram Singh

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PROFESSIONAL SUMMARY

Data / Business Analyst with 3 years of experience delivering analytics and process automation solutions for a Fortune 500 insurance client at Accenture. Skilled in SQL, Python, Power BI, and AWS RDS, with a track record of automating ETL workflows, designing dashboards, and translating business requirements into scalable analytical solutions. Focused on using data to improve operational efficiency and support business decision-making.

EDUCATION

University of Massachusetts, Amherst (Isenberg School of Management)

September 2024 - May 2026

Master of Science in Business Analytics

GPA: 3.9/4.0

Manipal University Jaipur, Rajasthan, India

July 2017 - July 2021

Bachelor of Technology in Electronics and Communication Engineering

EXPERIENCE

Business Architecture Analyst (Data & Analytics) , Accenture, India

June 2021 - July 2024

- Designed a 12-table AWS RDS (MySQL) database and defined Power Apps functional requirements to centralize form management workflows across three business units, reducing quarterly processing time by 40%.
- Audited and documented 3,000+ production forms across multiple insurance product lines, creating a centralized inventory that improved traceability and supported governance and change-management activities.
- Built Power BI and Tableau dashboards for 25+ business users across 5 departments, replacing manual Excel reports with automated reporting from AWS RDS to track workflow performance and turnaround times.
- Developed SQL queries using joins, CTEs, and window functions to analyze operational data, monitor KPIs, and track SLA compliance, contributing to an estimated \$300K in operational efficiency gains.
- Automated data extraction and validation using Python, reducing manual processing by 4–5 hours per reporting cycle while improving data consistency across SQL databases and shared files.
- Analyzed 50K+ insurance claims records quarterly in Python, identifying validation gaps affecting 15% of records and implementing automated data quality checks that improved reporting reliability.
- Worked with business and compliance teams to gather reporting requirements, document business rules, and translate stakeholder needs into technical specifications for developers.

PROJECTS

P&C Insurance Claims Analytics (SQL)

May 2026 - June 2026

- Designed a 7-table relational database modeling P&C insurance operations (policies, claims, payments, agents, and coverage lines) and developed advanced SQL queries using joins, CTEs, and window functions to analyze loss ratios, claim severity, settlement time, and regional claim frequency and severity.
- Built rule-based fraud scoring and customer risk segmentation logic using SQL CASE expressions and conditional aggregation to prioritize high-risk claims for SIU review and flag high-exposure policyholders.

Airbnb Review Sentiment & Pricing Analytics

February 2026 - May 2026

- Processed 320K Airbnb reviews into a listing-level dataset of 58.5K records using TF-IDF, n-gram extraction, and DistilBERT sentiment scoring, and compared LDA, NMF, and LSA topic models to extract guest experience themes from unstructured text.
- Engineered 85+ categorical, geospatial, and text-derived features and benchmarked Ridge Regression against a Keras neural network for price prediction, improving R^2 from 0.51 to 0.62, then used permutation importance and neighbourhood-level sentiment analysis to surface the strongest price drivers.

Customer Analytics & KPI Dashboard

February 2026 - May 2026

- Segmented \$500K+ e-commerce revenue across 8 categories and 4 channels, performing Recency-Frequency-Monetary (RFM) segmentation on 734 customers to identify 119 Champions (31.5% revenue) and 48 At-Risk high-value customers.
- Assessed marketing ROI across 6 advertising platforms, calculating Return on Ad Spend (ROAS) by platform and channel to recommend budget reallocation strategies for improved customer acquisition efficiency and profitability optimization.

ACHIEVEMENTS

Pinnacle Award (Q1'24) and Trailblazer Award (Q2'23), Accenture - recognized for contributions to analytics, data infrastructure, and process improvement.

TECHNICAL SKILLS

Technical Stack: SQL (PostgreSQL, SQL Server, MySQL), Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Power BI (DAX, Power Query), Tableau, Excel (Advanced), Power Apps.

Analytics & Methods: Data Visualization, EDA, A/B Testing, Cohort Analysis, RFM Segmentation, Statistical Analysis, Predictive Modeling (Regression, Forecasting), Data Validation, KPI Development, Trend Analysis, Root Cause Analysis.

Cloud & Business Tools: AWS RDS, ETL, Workflow Automation, Stakeholder Management, Agile/Jira, Requirements Gathering, Process Improvement, Data Modeling (relational schema design, star schema).